

Moore, IPS 6e Chapter 04

Quizzes prepared by Dr. Patricia Humphrey, Georgia Southern University

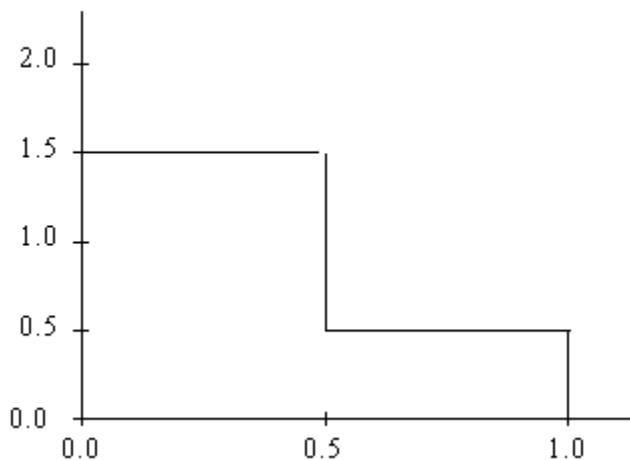
A college basketball player makes 80% of his free throws. At the beginning of a game he misses his first two free throws. We may correctly conclude

- ☐ a. He will make his next eight shots.
- ☐ b. He will make eight shots in a row sometime during the season but not necessarily in this game.
- ☐ c. None of the above.

A fair coin is tossed, and a fair six-sided die is rolled. Suppose the outcomes for the coin and die are independent. The probability of getting a head and rolling a six is

- ☐ a. $1/12$.
- ☐ b. $2/3$.
- ☐ c. $1/3$.

Suppose X is a continuous random variable taking values between 0 and 1 and having a probability distribution described by the following density curve.



The probability that X takes a value between 0 and $3/4$ is

- ☐ a. 0.875.
- ☐ b. 0.75.
- ☐ c. $3/8$.

The weight of a medium size orange selected at random from a large bin of oranges at the local supermarket is a random variable with mean $\mu = 12$ ounces and standard deviation $\sigma = 1.2$ ounces. Suppose we independently pick two oranges at random from the bin. The expected value of the sum of the weights of the two oranges, in pounds (1 pound = 16 ounces) is

- ☐ a. 24.
- ☐ b. 1.5.

☐ c. 0.75.

Two coins are tossed and the total number of heads is counted. Which of the following would be a legitimate probability model for the total number of heads? That is, which of the following models satisfies the rules of probability?

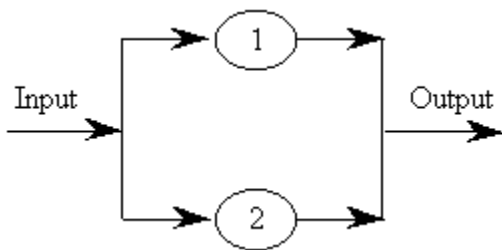
- ☐ a.

Number of heads:	0	1	2
Probability:	1/3	2/3	1/3
- ☐ b.

Number of heads:	0	1	2
Probability:	1/16	5/8	5/16
- ☐ c.

Number of heads:	0	1	2
Probability:	1/2	3/4	-1/4
-

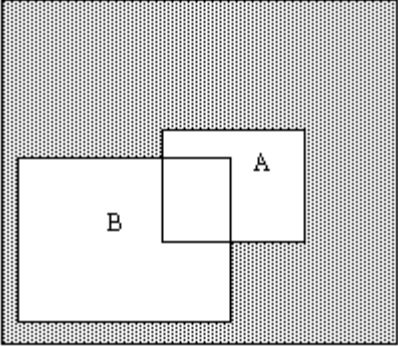
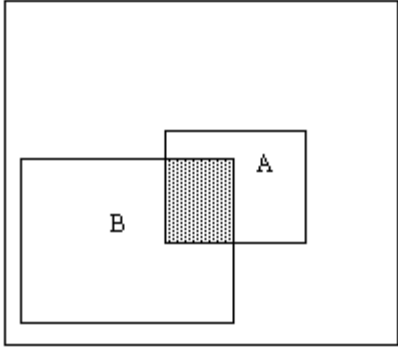
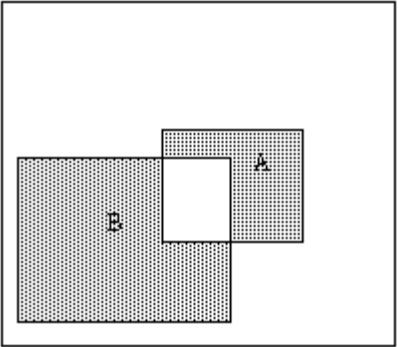
A system has two components that operate in parallel, as shown in the diagram below. Since the components operate in parallel, at least one of the components must function properly if the system is to function properly. The probabilities of *failure* for components 1 and 2 during one period of operation are .20 and .03, respectively. Let F denote the event that the component 1 *fails* during one period of operation and G denote the event that component 2 *fails* during one period of operation. The component failures are independent.



The event corresponding to the above system functioning properly during one period of operation is

- ☐ a. F or G
- ☐ b. F^C or G^C
- ☐ c. F^C and G^C
-

A plumbing contractor puts in bids in on two large jobs. Let the event that the contractor wins the first contract be A and the event that the contractor wins the second contract be B . Which of the following Venn diagrams has shaded the event that the contractor wins exactly one of the contracts?

- ☐ a. 
- ☐ b. 
- ☐ c. 

A fifth grade teacher gives homework every night in both mathematics and language arts. The time to complete the mathematics homework has a mean of 30 minutes and a standard deviation of 10 minutes. The time to complete the language arts assignment has a mean of 40 minutes and a standard deviation of 12 minutes. The time to complete the mathematics and the time to complete the language arts homework have a correlation $\rho = -0.3$. The mean time to complete the entire homework assignment.

- ☐ a. is less than 70 minutes since the negative correlation tells you that more time on one assignment will be associated with less time on the second assignment.
- ☐ b. is 70 minutes.
- ☐ c. is greater than 70 minutes since the measurements are correlated which raises the mean regardless of the sign of the correlation.

A fourth grade teacher gives homework every night in both mathematics and language arts. The time to complete the mathematics homework has a mean of 10 minutes and a standard deviation of 3 minutes. The time to complete the language arts assignment has a mean of 12 minutes and a standard deviation of 4 minutes. The time to complete the mathematics and the time to complete the language arts homework have a correlation $\rho = -0.75$. The standard deviation of the time to

complete the entire homework assignment is

- ☐ a. 25 minutes.
- ☐ b. 5 minutes.
- ☐ c. 4 minutes.

I toss a thumbtack 60 times and it lands point up on 35 of the tosses. The approximate probability of landing point up is

- ☐ a. 35.
- ☐ b. 0.35.
- ☐ c. 0.58.

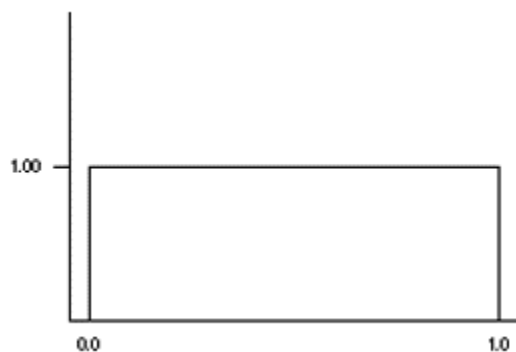
A game consists of drawing three cards at random from a deck of playing cards. You win \$3 for each red card that is drawn. It costs \$2 to play. For one play of this game, the sample space S for the net amount you win (after deducting the cost of play) is

- ☐ a. $S = \{\$0, \$1, \$2, \$3\}$
- ☐ b. $S = \{\$0, \$3, \$6, \$9\}$
- ☐ c. $S = \{-\$2, \$1, \$4, \$7\}$

In a particular game, a ball is randomly chosen from a box that contains 3 red balls, 1 green ball, and 6 blue balls. If a red ball is selected you win \$2, if a green ball is selected you win \$4, and if a blue ball is selected you win nothing. Let X be the amount that you win. The expected value of X is

- ☐ a. \$1
- ☐ b. \$2
- ☐ c. \$3

Let the random variable X be a random number with the uniform density curve given below.



$P(0.7 < X < 1.1)$ has value

- ☐ a. 0.30
- ☐ b. 0.40
- ☐ c. 0.70

Students at University X must be in one of the class ranks, Freshman, Sophomore, Junior, or Senior. At University X, 35% of the students are Freshmen and 30% are Sophomores. If a student is selected at random, the probability he or she is either a Junior or a Senior is

- ☐ a. 35%
- ☐ b. 65%
- ☐ c. 70%

Suppose there is a deck of three cards, one marked with a 1, one marked with a 2, and one marked with a 5. You draw two cards at random and without replacement from the deck of three cards. The sample space $S = \{(1, 2), (1, 5), (2, 5)\}$ consists of these three equally likely outcomes. Let X be the total of the two cards drawn. Which of the following is the correct set of probabilities for X ?

- ☐ a.

X	1	2	5
$P(X)$	1/3	1/3	1/3

- ☐ b.

X	3	6	7
$P(X)$	1/3	1/3	1/3

- ☐ c.

X	3	6	7
$P(X)$	3/16	6/16	7/16

Suppose we have a *loaded* die which gives the outcomes 1 through 6 according to the probability distribution

X	1	2	3	4	5	6

$P(X)$	0.1	0.2	0.3	0.2	0.1	0.1
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Note that for this die all outcomes are *not* equally likely as they would be if this were a fair die. If this die is rolled 6000 times, then $\bar{x}$, the sample mean of the number of spots on the 6000 rolls, should be about

- ☐ a. 3.00
- ☐ b. 3.30
- ☐ c. 3.50

Suppose we have a loaded die which gives the outcomes 1 through 6 according to the following probability distribution.

X	1	2	3	4	5	6
$P(X)$	0.1	0.2	0.3	0.2	0.1	0.1

Note that for this die all outcomes are not equally likely as they would be if this were a fair die. If this die is rolled 6000 times, the number of times we get a 2 or a 3 should be about

- ☐ a. 1000
- ☐ b. 2000
- ☐ c. 3000

Suppose we toss a penny and a nickel. Let A be the event that the penny is a head and B be the event that the nickel is a tail. The events A and B are

- ☐ a. disjoint
- ☐ b. complements
- ☐ c. independent

The weight of a medium-sized orange selected at random from a large bin of oranges at the local supermarket is a random variable with mean $\mu = 12$ ounces and standard deviation $\sigma = 1.2$ ounces. Suppose we independently pick two oranges at random from the bin. The difference in the weights of the two oranges selected (the weight of first orange minus the weight of the second orange) is a random variable with standard deviation

- ☐ a. 0.00 ounces

- ☐ b. 1.70 ounces
- ☐ c. 2.88 ounces

A process is called random if

- ☐ a. Individual outcomes are uncertain but happen in a predictable manner through time.
- ☐ b. One has no idea what will happen.
- ☐ c. Outcomes happen in a 50-50 split.

The weight of medium-sized oranges selected at random from a large bin of oranges at the local supermarket is a random variable with mean $\mu = 12$ ounces and standard deviation $\sigma = 1.2$ ounces. The weight of the oranges, in *pounds* (1 pound = 16 ounces) is a random variable with standard deviation

- ☐ a. 0.075 pounds
- ☐ b. 0.005625 pounds
- ☐ c. 19.2 pounds

A Psychology instructor asked the 100 females in her class to rate their intelligence on a scale of 1 to 10. Some of the ratings were

Rating	5	6	7	8	9	10
Number	24	18	38	12	4	4

What proportion of the students rated their intelligence less than 5?

- ☐ a. 6
- ☐ b. 0.06
- ☐ c. 0.94

A Psychology instructor asked the 100 females in her class to rate their intelligence on a scale of 1 to 10. The ratings were

Rating	5	6	7	8	9	10
Number	24	18	38	12	4	4

What proportion of the females rated their intelligence less than 7?

- ☐ a. 0.20
- ☐ b. 0.60
- ☐ c. 0.42.

Which of the following events are disjoint?

- ☐ a. Choose a student at random from a statistics class. Event A is that the student is a Junior.

Event B is that the student is a Senior.

- ☐ b. Choose a student at random from a statistics class. Event A is that the student is a psychology major. Event B is that the student is a Statistics major.
- ☐ c. Choose a student at random from a statistics class. Event A is that the student is at least 5'8" tall. Event B is that the student weighs at least 160 pounds.

On a certain airline, the chance the early flight from Atlanta to Chicago is full is 0.8. The chance the late flight is full is 0.7. The chance both flights are full is 0.6. Are the two flights being full independent events?

- ☐ a. Yes.
- ☐ b. No.
- ☐ c. We do not have enough information.

A Psychology instructor asked the 100 females in her class to rate their intelligence on a scale of 1 to 10. The ratings were

Rating	5	6	7	8	9	10
Number	24	18	38	12	4	4

What was the mean ranking for the females in this class?

- ☐ a. 7.5
- ☐ b. 6.66
- ☐ c. 612.

Females in a certain class rated their intelligence with a mean of 6.5. The males in the same class rated their intelligence with a mean of 7.25. The difference in the mean ratings (male - female) was

- ☐ a. -0.75 points
- ☐ b. 0.75 points
- ☐ c. 13.75 points

Females in a certain class rated their intelligence with a mean of 6.5 with standard deviation 1.12. The males in the same class rated their intelligence with a mean of 7.25, with standard deviation 1.50. The standard deviation of the difference in the ratings (male - female) was

- ☐ a. 0.38
- ☐ b. 3.50
- ☐ c. 1.87

Most people think babies are equally likely to come as either a boy or a girl. This is not true. Actually, about 51.3% of all babies are boys. If a family has two children (not twins), what is the chance both children are boys?

- ☐ a. 25%

- ☐ b. 102.6%
- ☐ c. 26.3%

Most people think babies are equally likely to come as either a boy or a girl. This is not true. Actually, about 51.3% of all babies are boys. If a family has two children (not twins), what is the chance they have one boy and one girl?

- ☐ a. 0.25
- ☐ b. 0.50
- ☐ c. 0.24

A Psychology instructor asked the 100 females in her class to rate their intelligence on a scale of 1 to 10. The ratings were

Rating	5	6	7	8	9	10
Number	24	18	38	12	4	4

The variance of the intelligence scores is

- ☐ a. 1.29
- ☐ b. 1.66
- ☐ c. 1.71

A refrigerator contains 6 apples, 5 oranges, 10 bananas, 3 pears, 7 peaches, 11 plums, and 2 mangos. Imagine you stick your hand into the refrigerator and pull out a piece of fruit at random. What is the sample space for this process?

- ☐ a. $S = \{\text{apple, orange, banana, pear, peach, plum, mango}\}$
- ☐ b. $S = \{6, 5, 10, 3, 7, 11, 2\}$
- ☐ c. $S = \{6 \text{ apples, } 5 \text{ oranges, } 10 \text{ bananas, } 3 \text{ pears, } 7 \text{ peaches, } 11 \text{ plums, } 2 \text{ mangos}\}$

A refrigerator contains 6 apples, 5 oranges, 10 bananas, 3 pears, 7 peaches, 11 plums, and 2 mangos. Imagine you stick your hand into the refrigerator and pull out a piece of fruit at random. What is the probability you pick a plum?

- ☐ a. $1/7$.
- ☐ b. $11/42$.
- ☐ c. $1/4$.

A loaded die is thrown. Which of the following is a legitimate probability distribution for this situation?

- ☐ a.

Value	1	2	3	4	5	6
Probability	0.1	0.2	0.2	0.2	0.3	0.0

- ☐ b.

Value	1	2	3	4	5	6
Probability	0.1	0.1	0.2	0.1	0.3	0.1

☐ c.

Value	1	2	3	4	5	6
Probability	0.3	-0.2	0.2	0.3	0.3	0.1

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